

Sri Hrushikesh Varma Bhupathiraju

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RESEARCH EXPERIENCE

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| Postdoctoral Research Associate
ASGUARD Lab, University of California, Irvine.
Advisor: Prof. Alfred Chen | Sept 2026 – Present |
| Graduate Student Researcher
Cyber-Physical Systems Security (CPSec) Lab, University of Florida.
Advisor: Prof. Sara Rampazzi | Jan 2021 – July 2026 |
| Research Intern
Toyota InfoTech Labs, California
Advisor: Dr. Michael Clifford | May 2025 – Aug 2025 |
| Graduate Student Researcher
Davis Cancer Research Center, Radiation Oncology, UF Health.
Advisor: Prof. Sanjiv Samant | Dec 2020 – Aug 2021 |
| Graduate Student Assistant
Modern Artificial Intelligence and Learning Technologies Lab, University of Florida.
Advisor: Prof. Sanjay Ranka | May 2020 – Aug 2020 |

RESEARCH AREAS

Cyber-physical system security & privacy, Autonomous systems, trustworthy AI, attack-aware perception and signal processing, critical infrastructure, and the Internet of Everything

EDUCATION

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| PhD in Computer Information Science and Engineering <i>University of Florida</i>
Research Area: Cyber-Physical Systems Security. Advisor: Prof. Sara Rampazzi | May 2026 |
| MS in Computer Information Science and Engineering <i>University of Florida</i>
Research Area: Medical Image Analysis. Advisor: Prof. Sanjiv Samant | May 2021 |
| B.Tech in Electronics and Communications Engineering <i>JNTU</i>
Research Area: Signal and Image Processing. Advisor: Prof. Sanjay Dubey | May 2019 |

PUBLICATIONS

* denotes equal contribution

- [1] **Sri Hrushikesh Varma Bhupathiraju**, Tetsu Ishizue, Nicholas Costagliola, Ozora Sako, Kentaro Yoshioka, Takeshi Sugawara, and Sara Rampazzi, “Illusion of Depth: Revealing Hidden Stereo Vision Vulnerabilities in Depth Estimation,” in *ACM Conference on Computer and Communications Security (CCS)*, 2026.
- [2] **Sri Hrushikesh Varma Bhupathiraju**, Shaoyuan Xie, Michael Clifford, Qi Alfred Chen, Takeshi Sugawara, and Sara Rampazzi, “The heat is on: Understanding and mitigating vulnerabilities of thermal image perception in autonomous systems,” in *Network and Distributed System Security (NDSS) Symposium*, 2026.
- [3] Jennifer Sheldon, Yungwoo Ko, **Sri Hrushikesh Varma Bhupathiraju**, Sara Osmanovic, Weidong Zhu, Md Jahidul Islam, and Sara Rampazzi, “Repliguard: Policy-driven replica management framework for protecting against acoustic attacks,” in *ACM SecDev*, 2026.

- [4] **Sri Hrushikesh Varma Bhupathiraju***, Arifu Onishi*, Rishikesh Bhatt, Sara Rampazzi, and Takeshi Sugawara, “Sound of interference: Electromagnetic eavesdropping attack on digital microphones using pulse density modulation,” in *USENIX Security Symposium*, 2025.
- [5] **Sri Hrushikesh Varma Bhupathiraju**, Takami Sato, Michael Clifford, Takeshi Sugawara, Qi Alfred Chen, and Sara Rampazzi, “To go or not to go: Shedding light on traffic light signal manipulation and defense strategies,” in *ACM Transactions on Cyber-Physical Systems*, 2025.
- [6] Daniel Olszewski, Allison Lu, Anna Crowder, Nathaniel Bennett, Seth Layton, **Sri Hrushikesh Varma Bhupathiraju**, Tyler Tucker, Siddhant Kalgutkar, Hunter Ver Helst, Carson Stillman, et al., “Reproducibility in applied security conferences: An 11-year review on artifacts and evaluation committees,” in *Proceedings of the 3rd ACM Conference on Reproducibility and Replicability*, 2025.
- [7] **Sri Hrushikesh Varma Bhupathiraju***, Takami Sato*, Michael Clifford, Takeshi Sugawara, Qi Alfred Chen, and Sara Rampazzi, “Invisible reflections: Leveraging infrared laser reflections to target traffic sign perception,” in *Network and Distributed System Security (NDSS) Symposium*, 2024.
- [8] **Sri Hrushikesh Varma Bhupathiraju**, Takami Sato, Michael Clifford, Takeshi Sugawara, Qi Alfred Chen, and Sara Rampazzi, “On the vulnerability of traffic light recognition systems to laser illumination attacks,” in *ISOC Symposium on Vehicle Security and Privacy (VehicleSec)*, ISOC, 2024.
- [9] Jennifer Sheldon, Weidong Zhu, Adnan Abdullah, **Sri Hrushikesh Varma Bhupathiraju**, Takeshi Sugawara, Kevin RB Butler, Md Jahidul Islam, and Sara Rampazzi, “Aquasonic: Acoustic manipulation of underwater data center operations and resource management,” in *IEEE Symposium on Security and Privacy (S&P)*, 2024.
- [10] **Sri Hrushikesh Varma Bhupathiraju**, Jennifer Sheldon, Luke A Bauer, Vincent Bindschaedler, Takeshi Sugawara, and Sara Rampazzi, “Emi-lidar: Uncovering vulnerabilities of lidar sensors in autonomous driving setting using electromagnetic interference,” in *ACM Conference on Security and Privacy in Wireless and Mobile Networks*, 2023.
- [11] **Sri Hrushikesh Varma Bhupathiraju***, Takami Sato*, Michael Clifford, Takeshi Sugawara, Qi Alfred Chen, and Sara Rampazzi, “Wip: Infrared laser reflection attack against traffic sign recognition systems,” in *ISOC Symposium on Vehicle Security and Privacy (VehicleSec)*, 2023.
- [12] Yulong Cao, **Sri Hrushikesh Varma Bhupathiraju**, Pirouz Naghavi, Takeshi Sugawara, Z Morley Mao, and Sara Rampazzi, “You can’t see me: Physical removal attacks on {lidar-based} autonomous vehicles driving frameworks,” in *USENIX security symposium*, 2023.
- [13] Jingxi Weng, Mohammad Anas, **Sri Hrushikesh Varma Bhupathiraju**, Thomas Samant, Alex Dresner, Jian Wu, and Sanjiv S Samant, “Organ-specific abdominal motion prediction using deep learning with cine-mri,” in *AAPM 65th Annual Meeting & Exhibition*, AAPM, 2023.
- [14] J Weng, **Sri Hrushikesh Varma Bhupathiraju**, T Samant, A Dresner, J Wu, and SS Samant, “Convolutional lstm model for cine image prediction of abdominal motion,” *Physics in Medicine & Biology*, vol. 69, no. 8, p. 085 024, 2023.
- [15] J Weng, **Sri Hrushikesh Varma Bhupathiraju**, T Samant, A Dresner, J Wu, and S Samant, “Deep learning based cine-mr image prediction and analysis for abdominal motion,” in *MEDICAL PHYSICS*, WILEY 111 River St, Hoboken 07030-5774, NJ USA, vol. 49, 2022, E498–E498.

TEACHING EXPERIENCE

Cyber-Physical System Security, Graduate Teaching Assistant Fall 2022
CISE Department, University of Florida. Supervisor: Dr. Sara Rampazzi

- Assisted a class of 90 students in a course on Cyber-Physical Systems and Security.
- Provided lectures on physical adversarial example modeling and training, and assisted students in simulating physical adversaries and testing as part of the lecture.
- Received strong positive feedback on Gatorevals, the university course evaluation system.

Digital Signal Processing, Undergraduate Teaching Assistant

Spring 2019

ECE Department, BVRIT, JNTU. Supervisor: Dr. Syed Audhagir Umar

- Instructed a class of 60 students, covering fundamental concepts such as denoising and image reconstruction.
- Guided hand-on projects helping students build real-world techniques such as image in-painting and object detection.

ACHIEVEMENTS

- Gator Engineering Award in Professional Excellence, 2026.
- Gartner Group Fellowship Award, University of Florida, 2025.
- Best Paper Award, ACM REP 2025.
- CSAW Applied Research Competition finalist, 2024 (Top 7.5%).
- ETAS Best Short Paper Award, VehicleSec '23.
- DENSO Best Demo Award, VehicleSec '23.
- Student Travel Grant Award, USENIX '23.
- Student Travel Grant Award, ACM WiSec '23.

SERVICES

Program Committee Member

- Lightning Talk and Tutorial Co-Chair, VehicleSec 2027.
- ACM AsiaCCS 2027.
- VehicleSec 2025, 2026.
- Poster and demo sessions, VehicleSec 2025.

External Reviewer

- IEEE Transactions on Dependable and Secure Computing Journal, 2025.
- IEEE Internet of Things Journal, 2025, 2026.
- USENIX Security, 2023, 2024.